Flexagons

Paper that hides its faces

A hands-on introduction for students

November 3, 2026

Take a strip of paper, fold it a few times, glue the ends together, and you hold a small machine. Pinch it, open it, and a picture that was nowhere to be seen appears. Pinch again, and it is gone. This booklet shows you how to make such *flexagons*, what to play with once you have them, who discovered them and why mathematicians still find them interesting. It accompanies the *Hexaflexagon Explorer* website, where you can flex a virtual flexagon and print templates.

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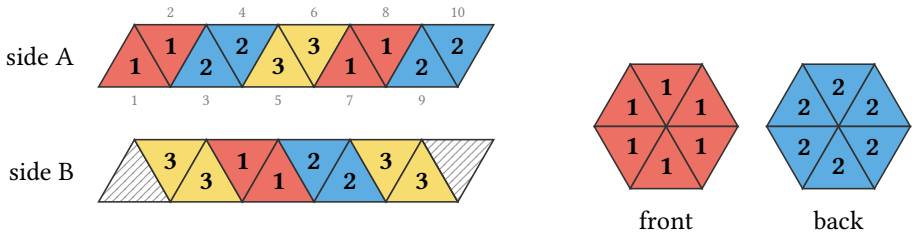
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(a) the strip: ten triangles, numbered on both sides (b) the finished trihexaflexagon

Figure 1: The trihexaflexagon. (a) A strip of ten equilateral triangles. The numbers of side B are drawn exactly behind the triangle that carries them; the hatched triangles are the glue tabs. The small grey numbers count the triangles. (b) After folding, one side of the hexagon shows six 1s, the other six 2s. The 3s have disappeared inside.

1 Make your own flexagon

1.1 What you need

- A sheet of ordinary paper. Thin paper folds better than card.
- Scissors, a glue stick, a pencil and a few coloured pencils.
- Ten minutes and a flat table.

The easiest start is to print a template from the *Hexaflexagon Explorer* website: the triangles are already drawn and numbered. If you have no printer, draw the strip yourself. The triangles must be *equilateral*, with all three sides the same length: about 3 cm is a good size. A strip of such triangles is easy to draw on a sheet of paper with a ruler: draw two parallel lines 2.6 cm apart, mark every 3 cm along both lines (shifted by 1.5 cm on the second line), and join the marks in a zigzag.

1.2 The trihexaflexagon

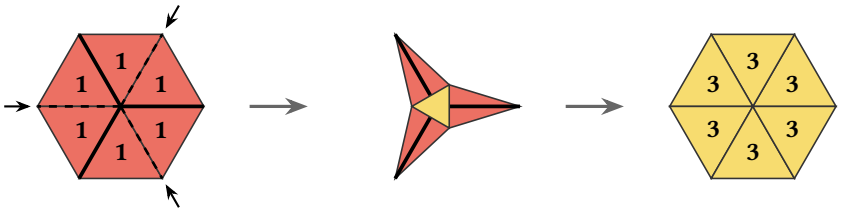
The name sounds complicated but says exactly what the object is. *Hexa* means six: the finished flexagon is a hexagon. *Tri* means three: it has three faces, although you can only ever see two of them.

Try it: fold a trihexaflexagon

1. Cut out a strip of ten equilateral triangles, Figure 1(a).
2. Number the triangles. Side A reads 1, 1, 2, 2, 3, 3, 1, 1, 2, 2. Turn the strip over along its long edge and number side B: glue, 3, 3, 1, 1, 2, 2, 3, 3, glue. Every number on side B sits exactly behind a triangle on side A.
3. Fold every line between two triangles forwards and backwards, so the strip becomes floppy. This is the most important step: a flexagon with stiff folds does not flex.
4. Put side A facing up. Fold the strip *behind* itself between the 2nd and the 3rd triangle. The two 3s on side B now touch each other.
5. Do the same between the 5th and the 6th triangle, and between the 8th and the 9th. Each time two 3s come face to face. The strip winds up into a hexagon.
6. The 10th triangle lands behind the 1st: glue tab on glue tab. Glue them together.

You should see a hexagon with six 1s on the front and six 2s on the back, Figure 1(b). The 3s are gone. Where are they?

The pinch flex. To find the 3s you have to *flex*. Hold the hexagon flat. Three of its six corners can be pushed down, the other three up, so that the hexagon folds into a three-pointed star, Figure 2. Pinch two



(a) pinch (b) open the centre (c) a new face

Figure 2: The pinch flex, schematically. (a) Three alternate radii (dashed) are folded inwards by pinching the adjacent triangles together, while the other three radii (bold) fold outwards. (b) The hexagon collapses into a three-pointed star, which opens up at the centre. (c) Flattening it again gives a hexagon that shows a face which was hidden before.

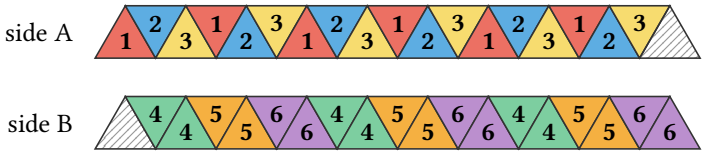


Figure 3: The strip of the hexahexaflexagon: nineteen triangles. Side A repeats 1, 2, 3; side B repeats 4, 4, 5, 5, 6, 6. The two glue tabs are at opposite ends.

neighbouring triangles together at one corner; do the same at the next-but-one corner, and at the one after that. Now look at the centre of the star: it can be opened from the top, like a flower. Open it and press the flexagon flat. A face full of 3s has appeared.

If the star will not open, nothing is broken. You pinched the wrong three corners. Flatten the flexagon, turn it by one corner and try again. Flexing takes a few tries at first and then becomes automatic.

Try it: watch the faces move

Write down which face is in front and which is at the back. Flex. Write it down again. Flex a third time. What do you notice? Keep flexing: does the face that was in front ever come back? Where did it go after one flex?

1.3 The hexahexaflexagon

The hexahexaflexagon has six faces (*hexa-hexa*). It is folded from a strip twice as long, with nineteen triangles, Figure 3. Side A repeats 1, 2, 3 and side B repeats 4, 4, 5, 5, 6, 6.

Try it: fold a hexahexaflexagon

1. Cut out and number the strip, Figure 3. Crease every line both ways.
2. Put side B up, with the glue tab at the left. Leave the glue tab alone. Fold the 2nd triangle onto the 3rd so the two 4s meet. Then fold the 4th onto the 5th: the two 5s meet. Go on like this to the end of the strip, always equal numbers of side B face to face. The strip coils up into a shorter strip of ten double triangles.
3. Keep the same side up. This short strip now looks like the one in Figure 1, and you fold it the same way: fold it behind itself between the 2nd and 3rd triangle so that two 2s meet, then between the 5th and 6th, then between the 8th and 9th.
4. Wrap the single triangle at the start over the last one, glue tab on glue tab, and glue.

The hexagon shows 1 on the front and 3 on the back. The other four faces are inside.

Flexing a hexahexaflexagon is more interesting than flexing a trihexaflexagon. Sometimes only one triple of corners works, sometimes two. Depending on where you pinch, different faces come out. Some

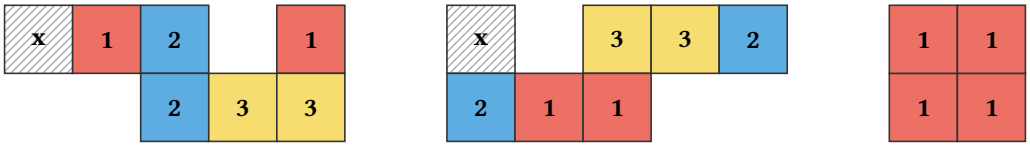


Figure 4: A flexagon made of squares: the tritetraflexagon. Left: the front of the seven-square sheet. Middle: its back, drawn as you see it after turning the sheet over from left to right. Right: folded, it is a 2×2 square showing four 1s on one side and four 2s on the other, with the 3s hidden.

faces are easy to find. Others seem to hide.

Game: the Tuckerman tour

Here is a recipe that is guaranteed to show you all six faces. Pick a corner. Flex there. If the flexagon opens, flex at the same corner again, and again, as long as it works. When it refuses, turn the flexagon by one corner and continue. This is called the *Tuckerman traverse*, after one of the people who first studied flexagons.

While you tour, keep a tally: every time a face appears, put a mark next to its number.

face	1	2	3	4	5	6
marks						

After twelve flexes you are back at the start. Which faces turned up most often? Which were rare? Compare with your friends' tallies.

1.4 A whole family of flexagons

More faces. Why stop at six? There are hexaflexagons with four faces, five faces, seven faces, and as many as you like. The strip just gets longer and the folding gets more fiddly. Some of them cannot be made from a straight strip at all: the strip has to zigzag or wind around itself before it is folded. There are even hexaflexagons whose strip overlaps itself so badly that it cannot be cut from a single flat sheet. With six faces there are already three different hexahexaflexagons. The one you made is the most common; the two others are folded from strips that are not straight.

Squares instead of triangles. A *tetraflexagon* is made of squares and is itself a square. The simplest one, the tritetraflexagon in Figure 4, has three faces, just like the trihexaflexagon, but it flexes differently: you fold it in half and open it from the other side, like a book that opens from its spine.

Try it: a flexagon made of squares

1. Copy the seven-square shape of Figure 4 onto squared paper (squares of 3 cm are good), cut it out and number both sides. Be careful with the back: the figure shows it after turning the sheet over from left to right.
2. Fold the square marked 2 in the top row *behind* its neighbour 1: the two 3s on the back now touch. The rest of the sheet swings round behind, to the left.
3. Fold the last 3 of the bottom row *under* the 3 next to it, so these two 3s touch too.
4. Now the two squares marked x lie in the same place. Tuck the one that came from the top left underneath the other, x on x, and glue.

You have a 2×2 square showing 1s on one side and 2s on the other. To flex: fold it in half, backwards, along the middle line. Then open it up from the middle. If it does not want to open, fold along the

other middle line instead, or turn it over. The 3s appear.

Other shapes and other flexes. Flexagons have been made from pentagons, from octagons, from triangles that are not equilateral, and from strips that spiral. People have also found new ways to flex the old flexagons. The *V-flex* of the hexahexaflexagon, for example, lifts two corners instead of three. It mixes the triangles of different faces, so that after a V-flex a face can be half 1 and half 4. The website shows the V-flex in 3D, and it is worth watching, because it is hard to believe the paper can do it until you see it.

1.5 Games and puzzles

None of these is hard. They are meant to be played, not solved.

Colour hunt

Colour every face of your hexahexaflexagon in a different colour, all six triangles the same. You will have to flex to reach the hidden faces. Then close your eyes, let a friend flex a few times, open your eyes and say which colour you expect to see next. Were you right?

The turning smiley

Draw a smiley face in the middle of one face of your trihexaflexagon, across all six triangles. Flex until the face comes back. Has the smiley changed? Now try the same on your hexahexaflexagon, on face 1 and on face 4. One of them will surprise you: the same face can come back in more than one version.

Message in a flexagon

Write a short message, one word per face, on the six faces of a hexahexaflexagon. Give it to a friend. Can they read the whole message? In which order do the words appear? Is there an order in which the message reads correctly?

Flexagon race

Everyone starts with their hexahexaflexagon showing face 1. On “go”, who is the first to show face 6? Then face 5? Afterwards, discuss: which face was the slowest to reach, and why?

Dots on the strip

Before folding a new trihexaflexagon, put a small dot in one corner of every triangle on both sides. After folding and flexing, look at a face: the six dots sit at the six corners, or at the centre, or somewhere in between. Flex, and look again. The dots show that the triangles do not just change colour, they also move.

Flexagon mazes

Figure 7(c) on page 9 is a map of the hexahexaflexagon. Every line is a position of the flexagon, and a flex moves you to another line of the same small triangle. Use the map as a maze: starting from the position showing 1 and 2, what is the smallest number of flexes to reach face 5? And from face 4 to face 6? Then check with the real flexagon. Depending on how it was folded, the faces 4, 5 and 6 of your own flexagon may sit at different tips of the map: find out which tip is which.

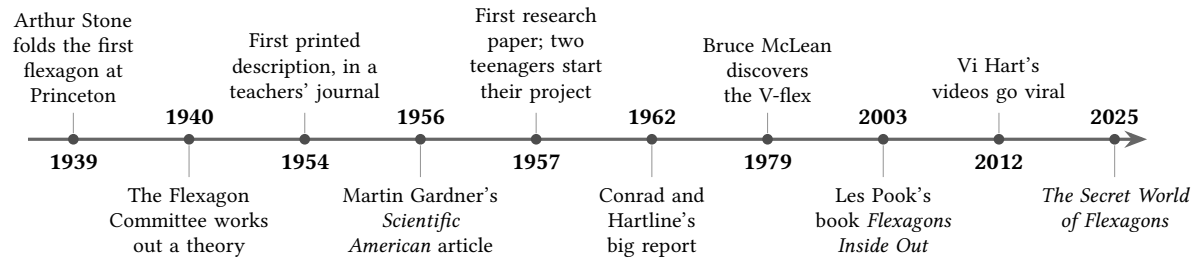


Figure 5: Some milestones in the history of flexagons.

2 A short history

Flexagons are young for a mathematical object. They were not discovered by a famous professor but by a student, by accident, while he was trying to make his paper fit in his folder.

2.1 An English student in America

In the autumn of 1939 Arthur Stone, a 23-year-old from London, arrived at Princeton University in the United States to study mathematics. He brought his English binder with him. American paper was wider than English paper, so he cut a strip off every sheet to make it fit. Soon he had a pile of long paper strips on his desk, and he started folding them to pass the time.

One of his folds turned a strip of triangles into a hexagon. He noticed that the hexagon could be pinched and opened, and that when he did this a face appeared that had been hidden inside. He had made the trihexaflexagon. The next day he worked out how to make one with six faces. He had found the hexahexaflexagon too.

2.2 The Flexagon Committee

Stone showed his paper toys to three friends at Princeton: Bryant Tuckerman, Richard Feynman and John Tukey. All four were young graduate students, and all four became well known later.

- **Arthur Stone** became a topologist, a mathematician who studies shapes that can be stretched and bent. You will meet topology in Section 3.
- **Bryant Tuckerman** worked on early computers. In 1971 he used one to find the largest prime number known at the time, a number with more than six thousand digits.
- **Richard Feynman** became one of the most famous physicists of the twentieth century and won the Nobel Prize in 1965.
- **John Tukey** became a leading statistician. He invented the word *bit* for a binary digit, and he was one of the first people to use the word *software*.

The four called themselves the *Flexagon Committee*. They met, folded, and tried to understand what they had. Tuckerman found the recipe for showing every face that you played with in Section 1.3: it is called the Tuckerman traverse to this day. Feynman drew diagrams of the possible flexes. By 1940 the committee had a complete theory of hexaflexagons.

They never published it. In 1941 the United States entered the Second World War, the four friends were scattered, and the theory stayed in their notebooks. For more than a decade flexagons were passed around by hand, from one person who had seen one to the next.

2.3 Flexagons go public

The first printed description appeared in 1954, in a short note in a British journal for mathematics teachers. It had the pleasant title *The flexagon and the hexahexaflexagram*, and it set off a small chain of replies from other teachers who had made their own.

The real breakthrough came in December 1956. Martin Gardner, a writer in New York with a love of puzzles and magic tricks, had heard about flexagons, travelled to Princeton, and interviewed Tukey and Tuckerman. His article *Flexagons* in the magazine *Scientific American* explained how to make them and told the story of the committee. Readers loved it. The magazine asked Gardner whether he could write something like it every month. He could: his column *Mathematical Games* ran for the next twenty-five years and introduced millions of people to mathematics as a game. It all started with a strip of paper.

2.4 The first papers, and two teenagers

In 1957 two mathematicians, Cletus Oakley and Robert Wisner, wrote the first research paper about flexagons. They gave a precise definition of the folded stacks of triangles, which they called *pats*, and showed how to count all possible hexaflexagons.

In the same year a stranger story began in Baltimore. Harold McIntosh, a mathematician at a research institute there, knew two high-school students who were about to graduate, Tony Conrad and Dan Hartline. He showed them flexagons and paid them for a summer project to study them. The teenagers did not stop at the end of the summer. Their reports, finished in 1960 and 1962, run to more than a hundred pages and remain one of the most complete accounts of how flexagons work. In 1961 Hartline wrote a computer program that flexes a flexagon, probably the first of its kind. Later Hartline became a neuroscientist, and Conrad became known as an experimental musician and film-maker. The flexagon reports were almost forgotten until they were put online more than forty years later.

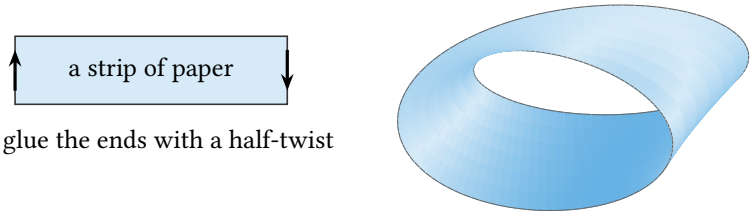
2.5 New flexes, new books, new audiences

Flexagons never went away. In 1979 Bruce McLean, a mathematician in Georgia, found that the hexahexaflexagon can be flexed in a completely new way, the V-flex, which reaches thousands of arrangements that the pinch flex never shows. Les Pook, a British engineer, collected everything that was known into two books, in 2003 and 2009, with dozens of flexagons to build.

In 2012 Vi Hart, a video maker, published a series of fast and funny hexaflexagon videos. They were watched many millions of times and taught a new generation to fold. Around the same time Scott Sherman started a website and a program that can simulate every flexagon and every flex anyone has thought of, and he kept discovering new ones. In 2025 Sherman, together with Yossi Elran and Ann Schwartz, published *The Secret World of Flexagons*, the newest book on the subject. Mathematicians still write papers about flexagons today, for instance about how to count them. Some of their questions are open: nobody knows a formula for the number of hexaflexagons that can be cut from a single sheet of paper.

Did you know?

Almost everybody in this story was young when they met flexagons: a 23-year-old student, his friends, two high-school graduates. You do not need a degree to discover something about flexagons. You need a strip of paper and a question.



(a) glue the ends with a half-twist

(b) a Möbius band

Figure 6: The Möbius band. (a) Glue the two short ends of a strip so that the arrows match: one end has to be turned over first. (b) The result has only one side and one edge.

3 The mathematics hiding inside

You have folded, flexed and played. What does any of this have to do with mathematics? Quite a lot. A flexagon is a meeting point of several mathematical fields, and each of them looks at it with different eyes. This section introduces four of them: topology, the Möbius band, graphs and groups. Each comes with something to try.

3.1 Topology: geometry without measuring

In school geometry you measure: lengths, angles, areas. *Topology* is the part of mathematics that deliberately does not. It asks which properties of a shape survive when you stretch it, squash it or bend it, as long as you do not cut it and do not glue new parts together. For a topologist a circle and a square are the same, because one can be bent into the other. A doughnut and a coffee mug are the same, because both have exactly one hole. A doughnut and a ball are different, because no amount of stretching makes a hole appear.

What does a topologist see in a flexagon? Not a hexagon. The hexagon shape comes and goes: during a flex the flexagon is a star, then a cup, then a hexagon again. What never changes is this: the flexagon is one long strip of paper whose ends are glued together. It is a *band*, a loop of paper. Flexing only moves the folds around; it never cuts the loop. So the questions a topologist asks are: what kind of loop is it, and how is it twisted?

Try it: unfold without cutting

Take a trihexaflexagon you do not mind spoiling. Open the folds gently, without tearing anything. You get a twisted loop of nine triangles. Count how many times the loop turns over as you follow it around once. That number is the topologist's description of the trihexaflexagon.

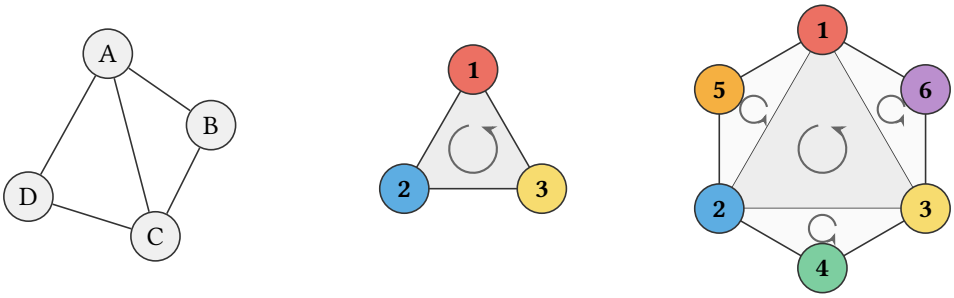
3.2 The Möbius band

The most famous twisted loop is the *Möbius band*, named after August Möbius, who described it in 1858. Take a strip of paper, turn one end over by a half-turn, and glue the ends together, Figure 6. The result has a strange property: it has only *one side*. If an ant walked along the band, it would come back to its starting point on the other side of the paper without ever crossing an edge.

Try it: the one-sided band

Make a Möbius band from a strip about 3 cm wide and 30 cm long.

- Draw a line along the middle of the band without lifting the pencil. Keep going until you reach the start. Look at the band: the line is on both sides of the paper.
- Now cut along the line. What do you expect to get? Two bands?
- Make another band, but this time turn the end over three half-turns before gluing. Draw the middle line again. One side or two?



(a) a graph (b) trihexaflexagon (c) hexahexaflexagon

Figure 7: (a) A graph: dots joined by lines. (b, c) The maps of our two flexagons, drawn as graphs. Each dot is a face. Each line joins the two faces that can be seen at the same time, one in front and one at the back. Going around a small triangle means flexing again and again at the same corners.

Here is the connection. The trihexaflexagon *is* a Möbius band, one with three half-twists instead of one. The strip you unfolded in the previous activity is exactly the three-half-twist band of the last experiment, pressed flat into a hexagon. So the trihexaflexagon is one-sided too.

Try it: draw through your flexagon

Take a trihexaflexagon and a pencil. Start in the middle of any triangle and draw a line into the next triangle of the strip, through the fold between them. Follow the paper wherever it leads, opening folds a little when you have to, and keep drawing. You will come back to the triangle you started on, but on the *other side* of the paper. Keep going until the line closes up. Now every triangle has a line on both its sides, and you drew it without ever lifting the pencil or crossing an edge.

The hexahexaflexagon is also a twisted band, with more twists. In fact the people who wrote the big 1962 report found that every hexaflexagon made from a straight strip is a Möbius-type band, and they could compute its number of twists from the number of faces.

3.3 Graphs: maps of what is possible

A *graph*, in this part of mathematics, is not a curve on squared paper. It is a collection of dots, called *vertices*, some of which are joined by lines, called *edges*, Figure 7(a). A map of the underground is a graph: stations are the vertices and the tracks are the edges. A friendship network is a graph. The rooms of a house, joined by doors, form a graph. Graphs are used to plan routes, to design networks and to search the web. Mathematicians began to study them in 1736, when Leonhard Euler asked whether one could walk through the city of Königsberg crossing each of its seven bridges exactly once. (One cannot.)

Flexagons come with a graph of their own. Take the faces as vertices. Join two faces by an edge if the flexagon can show them at the same time, one at the front and one at the back. Figure 7(b) shows the result for the trihexaflexagon: a single triangle, because every pair of faces can be seen together. For the hexahexaflexagon, Figure 7(c), the graph is a hexagon cut into four triangles. Faces 1, 2 and 3 sit in the middle and are joined to four other faces each. Faces 4, 5 and 6 sit at the tips and are joined to only two. This is why your tally from the Tuckerman tour was so uneven: the inner faces lie on more routes, so they turn up more often.

The graph is also the map for the maze puzzle in Section 1.5. A flex at the same corners walks you around one of the small triangles. To change triangles, you turn the flexagon by one corner. Once you have the map, you can plan a route from any face to any other.

Members of the Flexagon Committee drew such maps in 1940, and they are called *Tuckerman diagrams*. When the V-flex is allowed the graph grows enormously: a hexahexaflexagon then has 3420 different arrangements of its triangles, and the *Hexaflexagon Explorer* website draws the full graph of them.

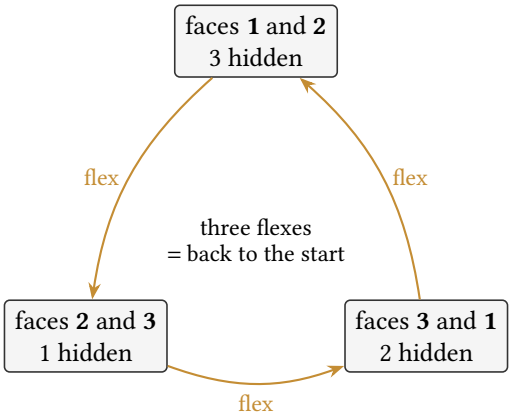


Figure 8: The trihexaflexagon has three positions. Each flex hides the face that was in front and brings out the hidden one, so three flexes in a row bring you back to where you started. Turning the flexagon over and flexing again runs the cycle backwards.

3.4 Groups: the algebra of moves

A *group* is the mathematician's way of describing a collection of moves that can be combined and undone. Think of the six ways to rotate a hexagon so that it looks the same: by 0°, 60°, 120°, 180°, 240° or 300°. Two rotations one after the other give another rotation: 120° followed by 300° is 420°, which is the same as 60°. Doing nothing, the rotation by 0°, is allowed. Every rotation can be undone: after 120°, rotate by 240° and you are back. These three rules, combining, doing nothing and undoing, are all that a group needs. Groups describe the symmetries of crystals, the moves of Rubik's cube and the way elementary particles can be exchanged.

Flexing is a move too. Figure 8 shows the three positions of the trihexaflexagon and the flex that takes one to the next. Each flex hides the face in front and reveals the hidden one, and after three flexes you are back where you started: in the language of groups, the flex has *order three*. Turning the flexagon over and flexing again undoes a flex, so it has an inverse. Together with the rotations of the hexagon and turning over, the flexes of the trihexaflexagon form a group, and mathematicians have worked out exactly which one it is.

For the hexahexaflexagon things are more subtle, because a flex is not always possible: at some corners the paper simply refuses. The moves still combine and can still be undone, but they are only defined in some positions. Mathematicians have a name for this kind of partial group as well, and the state graph of the previous section is the best way to picture it.

Try it: flexagon arithmetic

Starting from face 1 in front on your trihexaflexagon, do the following sequence of moves: flex, flex, turn over, flex, flex, flex, turn over, flex. Before you start, predict which face you will end on. Then check. Can you find a shorter sequence that ends in the same position?

3.5 Counting: how many flexagons are there?

One more field deserves a mention: *combinatorics*, the art of counting arrangements. Flexagons with three, four and five faces exist in one version each. With six faces there are three different ones, with seven faces three, with eight faces seven, with nine faces eight, and after that the numbers grow quickly. Behind these numbers hides the graph of the previous page: each hexaflexagon corresponds to a way of cutting a polygon into triangles, and ways of cutting polygons into triangles are counted by a famous sequence called the Catalan numbers. The same sequence counts the ways to put brackets into a sum, the paths through a grid that never cross the diagonal, and dozens of other things. Finding such hidden connections is what much of mathematics is about.

4 Think about it

Now that you have folded, flexed, read the history and met some of the mathematics, take a few minutes to think and write. There are no right answers. Discuss the questions with a friend, or write a short paragraph on each.

1. Is a flexagon a puzzle, a mathematical object or a machine? Explain your answer. Could it be all three at once?

2. What was the difficult part of flexing the object? Was it the paper, your fingers, or knowing which corners to pinch? What helped?

3. If you created a new flexagon, what would you change: the shape of the pieces, the number of faces, the way it flexes, or what is drawn on it? Sketch your idea.

4. How does colouring the faces help you to see the structure of the flexagon? What would you lose if all faces were white? What could you discover by colouring single triangles instead of whole faces?

5. The Tuckerman tour is a recipe that always works. Why do you think it works? Try to explain it using the map in Figure 7(c).

6. Arthur Stone discovered flexagons while playing with scraps of paper. Do you think playing is a serious way to do mathematics? Can you think of other discoveries that started as play?

7. The Flexagon Committee never published its theory, and two high-school graduates later rediscovered much of it on their own. What is lost when results are not written down? What is gained when someone has to find them again?

8. The trihexaflexagon is a Möbius band. Did that surprise you? What other everyday objects might secretly be twisted loops?

9. A graph of a flexagon tells you which faces can be reached from which. What other everyday situations could you describe with a graph of dots and lines?

10. Once you knew the map of the hexahexaflexagon, did flexing feel different? Is a trick still a trick once you understand it?

11. What question about flexagons would you most like to have answered? How would you start looking for the answer?

Hints for the puzzles

- **Watch the faces move.** After a flex, the face that was in front has folded away inside. The face that was at the back is still there, and the third face has come out to join it. After three flexes everything is back as it was.
- **Tuckerman tour.** In twelve flexes the faces 1, 2 and 3 appear three times each, the faces 4, 5 and 6 only once each.
- **The turning smiley.** On the trihexaflexagon the smiley comes back unchanged. On the hexahexaflexagon, face 1 can come back with its six triangles rearranged, so the smiley is cut up and turned. Face 4 always comes back the same.
- **Flexagon mazes.** In the map, every line is a position of the flexagon and every flex moves to a neighbouring line of the same small triangle. From a position showing 1 and 2 you can reach the tip face 5 in two flexes; from the tip 4 to the tip 6 you need at least three.

- **Flexagon arithmetic.** Write down the two visible faces after every move. A turn-over makes the next flexes run around the cycle the other way, so the long sequence can be replaced by a much shorter one.

Where to go next

- the *Hexaflexagon Explorer* website, with a 3D model of the pinch flex and the V-flex, the state graph, and printable templates for the trihexaflexagon and the hexahexaflexagon.
- Martin Gardner, *Hexaflexagons and Other Mathematical Diversions*, University of Chicago Press, 1988. The first chapter is the 1956 article that started it all.
- Vi Hart, *Hexaflexagons*, video series, 2012. Search for "Vi Hart hexaflexagons".
- Scott Sherman, *Explorable Flexagons*, `http://loki3.com/flex/`. Simulations, templates and a catalogue of flexes.
- Scott Sherman, Yossi Elran and Ann Schwartz, *The Secret World of Flexagons*, CRC Press, 2025.
- Les Pook, *Flexagons Inside Out*, Cambridge University Press, 2003.
- Antony Conrad and Daniel Hartline, *Flexagons*, 1962, online at `https://erikdemaine.org/flexagons/Conrad-Hartline-1962/`. The teenagers' report; harder reading, but a piece of history.
- Marshall Hampton, *Constructing and counting hexaflexagons*, arXiv:1704.07775, 2017. For readers who want to see how the counting is done.